Bluestock Mutual Fund Analytics

Capstone Project Final Report

Prepared by: Data Engineering & Analytics Team

1. Executive Summary

This report outlines the end-to-end implementation of the Bluestock Mutual Fund Analytics Platform. The project consolidates fragmented mutual fund data into a centralized, production-ready SQLite data warehouse. It includes robust data engineering, automated ETL pipelines, performance & risk analytics, and an interactive Streamlit dashboard.

• Total Funds Tracked: 40
• Industry AUM: ■62.74 Lakh Crore
• Average 3Y CAGR: 14.09%

2. Architecture & Data Engineering

The system uses a classic ETL architecture. Python and pandas handle extraction from 10 raw CSV datasets. Transformations include date parsing, missing value forward-filling (ffill for NAVs on weekends/holidays), and metric computations. The loaded data is structured in a 10-table Star Schema within SQLite.

3. Top Performing Funds

Scheme Name	Category	AMC	3Y Return	Sharpe Ratio
SBI Small Cap Fund - Regular P...	Small Cap	SBI Mutual Fund	23.39%	0.94
SBI Small Cap Fund - Direct Pl...	Small Cap	SBI Mutual Fund	23.14%	0.93
ABSL Small Cap Fund - Regular ...	Small Cap	Aditya Birla Sun Life MF	22.38%	0.90
Axis Small Cap Fund - Regular ...	Small Cap	Axis Mutual Fund	20.98%	0.84
Nippon India Small Cap Fund - ...	Small Cap	Nippon India MF	20.15%	0.81

4. Advanced Analytics & Machine Learning

• **K-Means Clustering:** Grouped 40 funds into 3 clusters based on risk-return profiles.
• **Monte Carlo Simulation:** Projected 5-year NAV growth paths using historical volatility.
• **Recommendation Engine:** Developed a content-based filtering system matching investor risk appetite to optimal funds.
• **Markowitz Efficient Frontier:** Calculated optimal portfolio weights for 5 selected equity funds.

5. Limitations & Future Scope

• **Limitations:** Static historical data dependency. AUM data is quarterly, limiting daily correlation analysis.
• **Future Scope:** Integration of real-time market news sentiment analysis, deployment of the database to PostgreSQL on AWS/GCP, and building user-specific portfolio tracking in the Streamlit app.